

SOC Report: Firewall Log Analysis

Objective

The goal is to analyze simulated firewall logs in Splunk to identify security threats such as:

- High-risk port access attempts
- Repeated access attempts
- Intrusion detection system (IDS) alerts
- Unusual traffic patterns

The log file `firewall_logs.txt` was ingested into Splunk using the **Upload** method, and indexed under `firewall_logs`.

Each log event contains the following fields:

- `timestamp`
- `action` (ALLOW or BLOCK)
- `src` (source IP)
- `dst` (destination IP)
- `port` (destination port)
- `protocol` (TCP/UDP)
- `reason` (e.g., Intrusion detection system alert, Normal traffic)

Field Extraction

Since the log format is structured text, the following **SPL field extraction** command can be used to extract key fields:

![Screenshot of the Splunk Search interface showing a new search for firewall logs. The search bar contains a regex query: | rex]((?<src_ip>\d{1,3}(?:\.\d{1,3}){3}))

New Search

1 | index=firewall_logs | rex "src=(?<src_ip>\d{1,3}(?:\.\d{1,3}){3}) dst=(?<dst_ip>\d{1,3}(?:\.\d{1,3}){3}) port=(?<port>\d+) protocol=(?<protocol>\w+) reason=(?<reason>.+)" |

✓ 100 events (before 25/03/2025 18:00:12.000) No Event Sampling

Events (100) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 minute per column

List Format 20 Per Page

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS		>	25/02/2024 19:18:54.000	2024-02-25 19:18:54 ALLOW src=174.244.162.197 dst=146.139.122.38 port=44998 protocol=UDP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
a host 1		>	25/02/2024 19:17:54.000	2024-02-25 19:17:54 ALLOW src=27.35.182.29 dst=25.154.25.67 port=8642 protocol=TCP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
a source 1		>	25/02/2024 19:17:54.000	2024-02-25 19:17:54 ALLOW src=36.92.160.158 dst=188.13.254.73 port=21732 protocol=UDP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
a sourcetype 1		>	25/02/2024 19:16:54.000	2024-02-25 19:16:54 ALLOW src=180.37.160.1 dst=23.172.24.215 port=7950 protocol=UDP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
INTERESTING FIELDS		>	25/02/2024 19:16:54.000	2024-02-25 19:16:54 ALLOW src=91.58.188.254 dst=159.13.249.236 port=14668 protocol=TCP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
# date_hour 2		>	25/02/2024 19:16:54.000	2024-02-25 19:16:54 ALLOW src=128.62.17.85 dst=0.60.132.43 port=37373 protocol=TCP reason=Normal traffic host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
# date_mday 1		>	25/02/2024 19:15:54.000	2024-02-25 19:15:54 BLOCK src=40.237.58.89 dst=179.48.61.101 port=443 protocol=TCP reason=High-risk port access attempt host = Pradeepreddys-MacBook-Pro.local source = firewall_log_example-1.txt sourcetype = firewall logs
# date_minute 53				
# date_month 1				
# date_second 1				
# date_wday 1				
# date_year 1				
a date_zone 1				
a dst_ip 100				
a index 1				
# linecount 1				
# port 50				

```
| rex "src=(?<src_ip>\d{1,3}(?:\.\d{1,3}){3})
dst=(?<dst_ip>\d{1,3}(?:\.\d{1,3}){3}) port=(?<port>\d+)
protocol=(?<protocol>\w+) reason=(?<reason>.+)"
```

This extracts:

- **src_ip** — Source IP address
- **dst_ip** — Destination IP address
- **port** — Destination port
- **protocol** — TCP or UDP
- **reason** — Reason for ALLOW/BLOCK

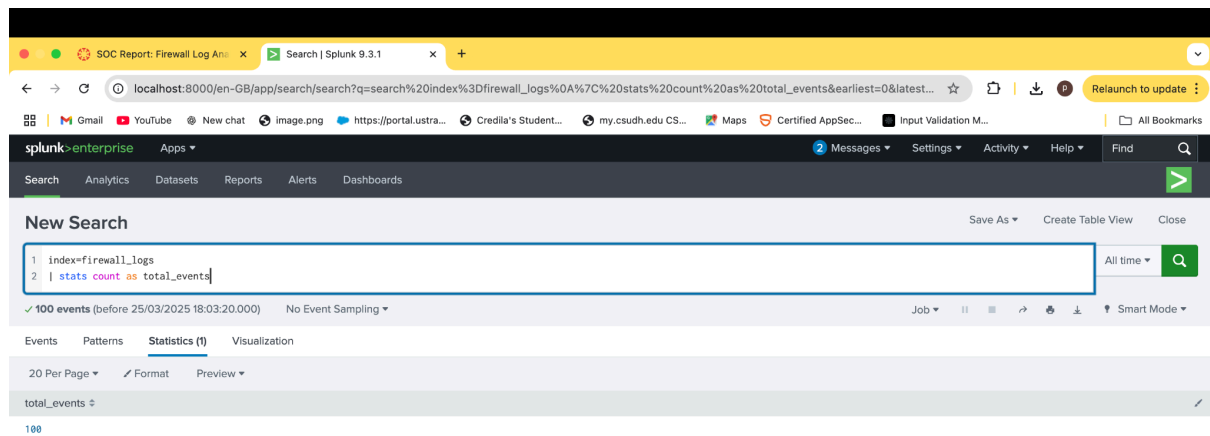
Data Analysis

1. Total Events Analyzed

Get a count of all events in the index:

```
index=firewall_logs
```

```
| stats count as total_events
```



2. High-Risk Port Access Attempts

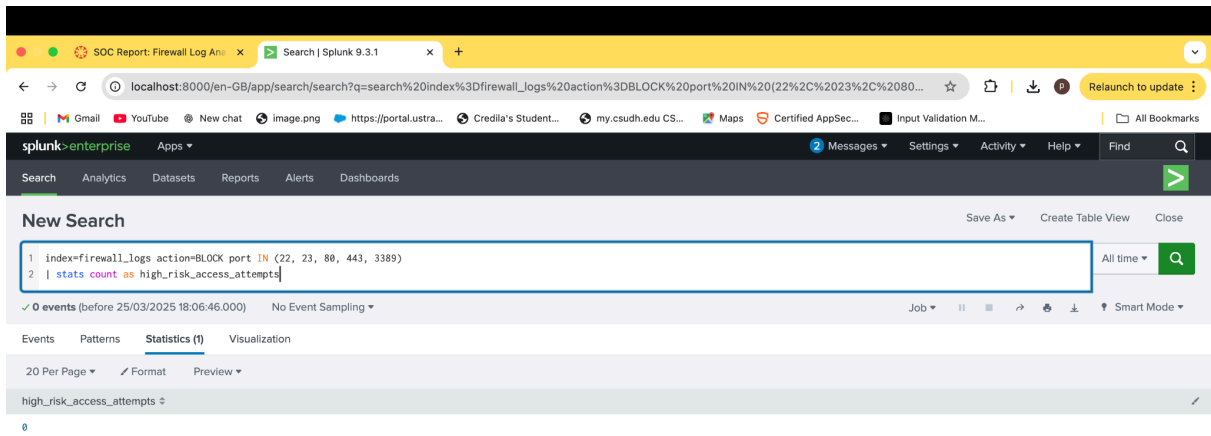
Focus on blocked events on sensitive ports (22, 23, 80, 443, 3389):

```
index=firewall_logs action=BLOCK port IN (22, 23, 80, 443, 3389)
```

```
| stats count as high_risk_access_attempts
```

```
index=firewall_logs action=BLOCK port IN (22, 23, 80, 443, 3389)
```

```
| table _time src_ip dst_ip port protocol reason
```

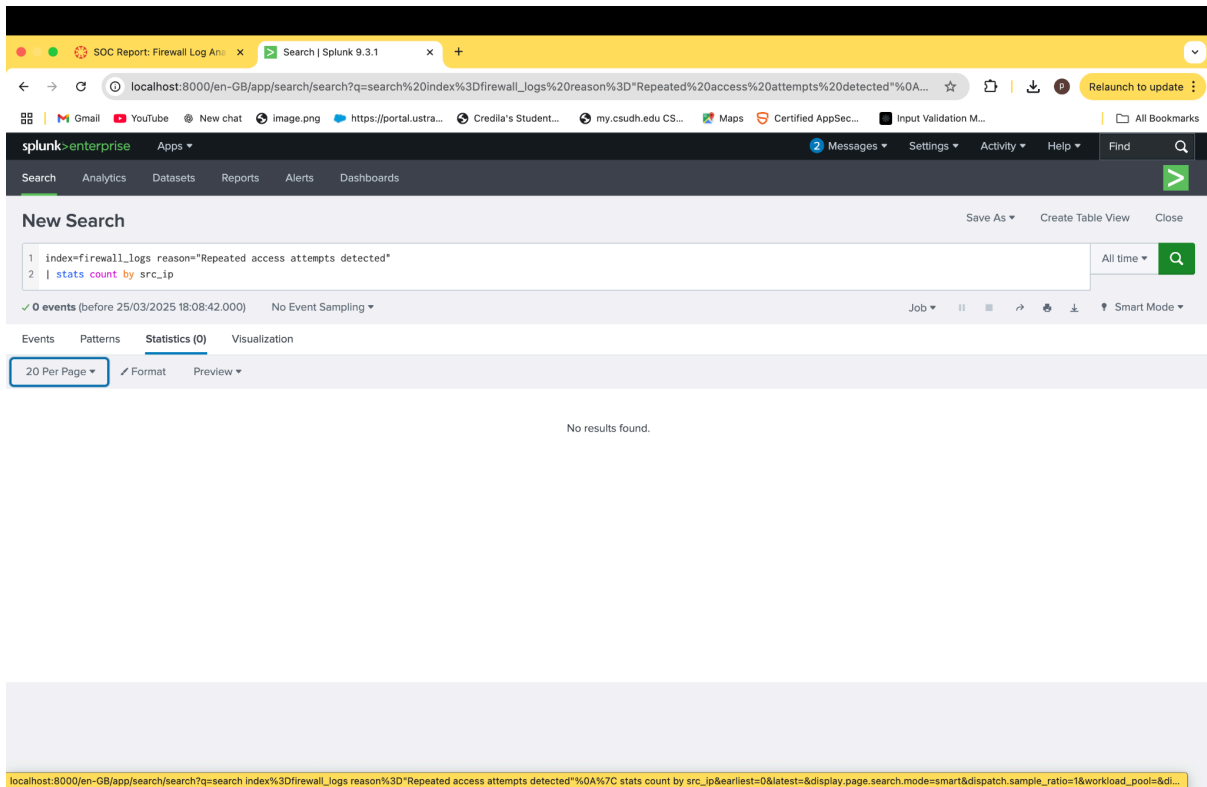


3. Repeated Access Attempts

Repeated access attempts may suggest brute-force attempts:

```
index=firewall_logs reason="Repeated access attempts detected"
```

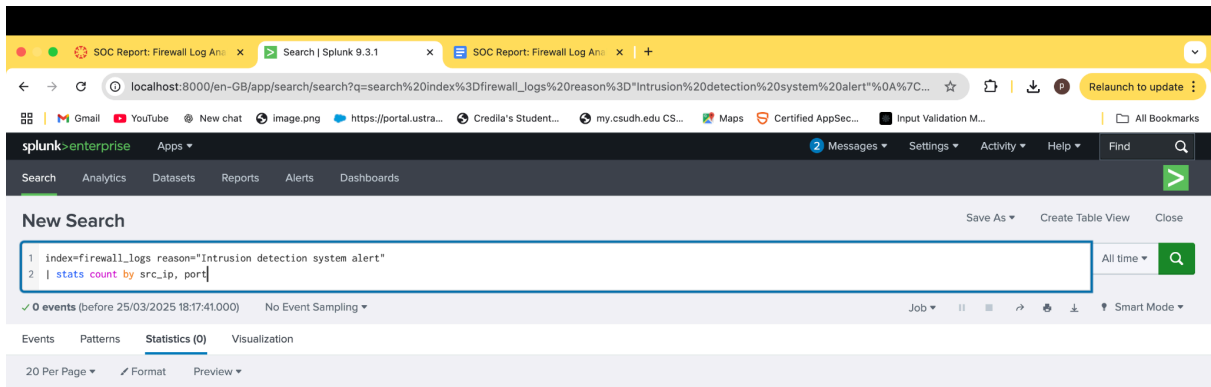
```
| stats count by src_ip
```



4. Intrusion Detection Alerts

Identify logs triggered by intrusion detection system:

```
index=firewall_logs reason="Intrusion detection system alert"
| stats count as intrusion_alerts
index=firewall_logs reason="Intrusion detection system alert"
| table _time src_ip dst_ip port protocol
```



5. Blocked Traffic Over Time

Analyze blocked traffic volume by hour:

```
index=firewall_logs action=BLOCK  
  
| bin _time span=1h  
  
| stats count as blocked_count by _time  
  
| sort - blocked_count
```

SOC Report: Firewall Log Analysis | Search | Splunk 9.3.1

localhost:8000/en-GB/app/search/search?q=search%20index%3Dfirewall_logs%20action%3DBLOCK%0A%7C%20bin%20_time%20span%3D1h%0A...

Relaunch to update

splunk>enterpriseApps

MessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboards

New Search

Save AsCreate Table ViewClose

1 index=firewall_logs action=BLOCK
2 | bin _time span=1h
3 | stats count as blocked_events by _time
4 | sort -blocked_events

All time

0 events (before 25/03/2025 18:16:11.000)No Event Sampling

Job

EventsPatternsStatistics (0)Visualization

20 Per PageFormatPreview

No results found.